

ocm_progress_report

OCM Project — Progress Report

From: Parth Nagar **To:** Dr. M S Srinath **Date:** 7 September 2026 **Covers:** all work from the start of the project to today, including everything done after Prof. Taniike's review

1. What we set out to do

We predict C₂ yield for OCM catalysts from their composition.

We hold two datasets. The **lab data** has 89,074 measurements from 2025. The **literature data** has 3,852 measurements published between 1982 and 2019.

We asked one question. Can the literature data make the lab model more accurate?

Two problems make this hard.

- **Label shift.** Published papers report higher yields than the lab sees. The gap is 3.4 percentage points.
 - **Covariate shift.** The two datasets use different chemistry.
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2. The five methods we tried first

We tried five methods, in this order. Each one answered a problem the previous one exposed.

#	Method	What it does	Result
1	Baseline	Train LightGBM on lab data only.	The number to beat.
2	Direct merge	Pool lab and literature data. Train one model.	Worse. The label shift pulls predictions up.
3	Selective merge (DRST)	Keep only literature that looks like lab chemistry. Then merge.	No real gain.

#	Method	What it does	Result
4	Selective merge (KMM)	Weight each literature record instead of filtering. Then merge.	Worse.
5	PFT (our method)	Train a model on literature. Feed its <i>prediction</i> into the lab model as an extra input. Never use literature yields as training labels.	Best result at the time.

Methods 2, 3 and 4 all failed for the same reason. They all put literature yields into the training target. The label shift then biases the model.

Method 5 avoided that. It passed literature knowledge as a *feature*, not a *label*. We reported a 10.6% improvement. We sent this to Prof. Taniike.

3. What Prof. Taniike found

He replied with six technical points. One of them changed the project.

His main point: our test was not measuring what we claimed.

The lab data holds 89,074 rows. But it only holds **917 distinct catalysts**. Each catalyst was measured about 97 times under different conditions.

We split the data randomly by row. So the same catalyst appeared in both training and testing. The model had already seen the catalyst it was being tested on.

He was right.

4. What we did about each of his six points

Point 1 and 2 — Split by catalyst, not by row

We rebuilt the evaluation. Now every measurement of a catalyst stays in one fold.

The result reversed:

Split method	Baseline	Our method (PFT)
By row (what we sent him)	2.118	1.912 — 9.7% better
By catalyst (correct)	2.943	2.995 — 1.8% worse

Our improvement disappeared. We withdrew the claim.

We also found the cause. Our Stage-1 model trained on literature *plus lab training rows*. Under a row split, those rows included the test catalysts. So the feature was carrying memorised answers, not literature chemistry.

We made catalyst-based splitting the default in all our code. It cannot be skipped by accident now.

Point 3 — Hold out whole catalyst families

We did this for 28 element families.

We found the Ba family fails badly. We then proved why. When no Ba catalysts appear in training, the model cannot use the Ba column at all. We deleted the column and got identical predictions. That proves the model ignores it.

Ba matters because it holds 78% of the lab’s best catalysts. Remove Ba, and you remove the high-yield chemistry from training.

We then asked a more useful question. **How many catalysts of a new family must we measure before the model can predict it?**

The answer depends on how you define “predictable enough”, and the two sensible definitions differ by a factor of five. For Ba the curve runs 0.509 with no Ba in training, 0.565 after 10 Ba catalysts, 0.616 after 25, 0.651 after 50, and 0.683 with all 204.

Our stored number, **10**, comes from asking when the model reaches 80% of its *final score*. But Ba already reaches 0.509 having seen no Ba at all, so that bar is nearly cleared by seeing nothing. If we instead ask when the model captures 80% of the *improvement* that seeing Ba brings, the answer is about **50**. We now think 50 is the more honest figure, and we report both.

The same caution applies to La, Ti, Zr and Ce, where our stored answer is “none needed”. That says more about how lenient the first definition is than about those families.

Point 4 — Predict the best yield per catalyst, not the best condition

He explained the real goal. His lab tests every catalyst under a fixed set of conditions. So the model does not need to pick the condition. It needs to say whether a catalyst can reach a high yield at all.

We rebuilt the model to predict each catalyst’s **maximum** yield. It now trains on 917 rows instead of 89,074. It ranks just as well.

Point 5 — Use catalyst-level metrics

We replaced point-wise RMSE with ranking metrics. These are what a screening campaign actually cares about.

Metric	All 917 catalysts	95% range	Equal-effort set (771)
Rank correlation of predicted vs real	0.761	0.725 – 0.785	0.724

Metric	All 917 catalysts	95% range	Equal-effort set (771)
best yield			
Enrichment of top performers	4.28×	3.04 – 4.89×	3.77×
Hit rate in a 20-catalyst shortlist	0.44	0.15 – 0.65	0.35

We report ranges, not single numbers. The ranges are wide, and we say so.

The last column is new, and it lowers our own score. The lab did not run every catalyst equally. Catalysts that were tested more thoroughly also tend to look better. So part of our score could be a record of which experiments the lab decided to finish, rather than chemistry. We checked how much.

The “equal-effort set” is the 771 catalysts that got at least 20 measurements at one temperature. On that set, the link between how much a catalyst was measured and how good it looks disappears: the correlation falls from 0.293 to 0.003. Our score on that set is lower. We report both numbers.

We also ruled out the obvious objection, that the drop is just from scoring fewer catalysts. We drew 300 random groups of 771 from the same predictions. Those score 0.767, in a range of 0.756 to 0.780. Our equal-effort score of 0.724 is below that range. So the drop is real.

The test we trust most. We trained the same model to predict *how many times a catalyst was measured*. It never saw a yield. Ranking catalysts by that alone gives a rank correlation of 0.400 with real best yield — but it finds top performers at 0.87×, which is no better than picking at random. So rank correlation can be partly bought with experimental effort. Enrichment cannot. That is why we treat enrichment as our main metric.

One limit this exposes. Among the catalysts the model ranks highest, its internal ordering is close to meaningless: rank correlation inside its top 150 picks is 0.179, and inside its top 20 it is −0.066. The model picks a good set — its top 20 average 17.3% best yield against about 10.5% for the whole library — but it cannot tell you which one inside that set is best. A shortlist should be treated as a group to test, not as a ranking to trust.

Point 6 — Test whether the rescaling step matters

He suspected our quantile-normalisation step did nothing. Tree models read the *order* of a feature, not its scale.

We tested three versions: normalised, raw yields, and plain ranks. The differences run from 0.001 to 0.023 RMSE. At three seeds, that sits inside run-to-run noise.

So we cannot separate them. We can drop the normalisation step without a measurable penalty. He was right.

5. What we tried after that, to rescue the method

The leak explained the old result. It did not prove literature data is useless. So we tested four honest ways to use it.

Design	Rank correlation	vs. control
Composition only (control)	0.761	—
Literature rank prior	0.757	−0.004
Similarity-to-literature features	0.740	−0.021
Both combined	0.744	−0.017
Catalyst-level direct merge	0.758	−0.003

We fixed the success criteria before running. **None of them beat composition alone.**

We also tested one idea of our own. One family (Zr) suggested literature helps where our own data is thin. We tested that across all 28 families. The effect vanished. We discarded our own idea.

6. The one place literature data does help

We then asked a different question. Can the lab model predict catalysts made by *other* preparation methods?

Test	Rank correlation
Literature made by impregnation (same method as lab)	0.398
Literature made by other methods	0.238
Other methods, after adding impregnation literature to training	0.388

Adding literature raised performance by 0.150, on every one of five seeds.

This is the first time literature data measurably helped. The reason is simple. Our lab only uses impregnation. So for other preparation methods, the lab has no data at all, and the literature is the only information we have.

Two honest caveats. Absolute performance is still poor. And plain merging beat our two-stage method here.

7. Where we stand today

What works. The composition-only model ranks unseen catalysts usefully. A campaign guided by it finds good catalysts about 3 to 5 times faster than random choice.

What does not work. Our PFT method does not improve prediction for lab catalysts. No literature-integration design does.

What we cannot do. The model cannot price a promoter family it has never seen. This is structural, not a coding problem.

What we produced. We enumerated 26,414 candidate catalysts using the lab's own design rules. We recommend a 17-catalyst campaign: 12 at the model's best predictions, and 5 spanning alternative supports to test whether the model's preference for Ba is real chemistry.

8. A correction we made to our own report

We audited our own work note before sending it. We found eight factual errors and fixed them all.

The most serious ones:

1. We misquoted what version 1 said. Version 1 reported 1.907. Our draft said 1.912.
2. We claimed the rescaling ablation differed by “0.001–0.005 RMSE”. The true range is 0.001–0.023.
3. We claimed Ba is uniquely hard to predict. Our own data shows five families score worse.
4. We quoted a data budget that contradicted our own stored measurement.
5. We presented an ensemble spread (± 0.09) next to a predicted yield. A reader would take it as an error bar. The real error is about 2.7 yield points.

We caught all of these before sending anything. Every number in the corrected note now traces to a stored file.

9. A second check, done afterwards

We then checked every number in our own handover document against the data itself. We did not trust the document. We recomputed each figure from the raw file, or re-read it from the stored result files.

Most numbers held. Every headline figure came out exactly right. We also re-ran two whole experiments from scratch. Both reproduced the stored results digit for digit, even though the software libraries had moved on several versions in between.

Three numbers had no source at all. No script computed them. Two of the three were wrong.

1. **A measurement-budget figure.** We had told Prof. Taniike that measuring 20 runs per catalyst instead of 135 reproduces the full ranking at 0.955. Recomputing it gives **0.949**. The 0.955 figure belongs to a 25-run plan, not a 20-run one. We also found the plan only applies to 759 catalysts, not the 811 we quoted. All three numbers are corrected in the note.
2. **A test about the 27 settings.** We had claimed a statistical signature proved the 27 rows in each cell are a complete set rather than a truncated one. No script ever computed that signature. The conclusion is right, but we had missed a much simpler proof: **104 cells hold more than 27 rows**, which a “keep only the top 27” export could not produce. We now use the count.

A discovery about the file itself. We wanted to know whether the 27 rows in each cell are 27 different reaction conditions, or 27 samples taken over time from a single run. This matters. If it is time, then a catalyst's best number is what it does when fresh, not what it sustains. Two earlier attempts to settle it failed. We now know why. **The file stores every cell sorted from highest yield to lowest.** Row order records rank, not the order in which the measurements were taken. So no further analysis of this file can answer the question. Only the lab can. The work note already asks.

We bounded the risk instead. We tested what would change if the question were answered the way we did not expect. We ranked catalysts by their worst measurement rather than their best. The two rankings do disagree at the top: only 7 of the top 20 catalysts are the same. But a model trained on the best measurement still ranks catalysts nearly as well when judged against the worst. It loses 0.017 in rank correlation, which is less than we gain by simply averaging over random seeds. It still finds good catalysts about 4 times faster than random choice, whichever way we judge it.

The answer changes the labels. It does not change which catalysts we would tell the lab to make. So this question is worth asking, but the campaign does not have to wait for the reply.

10. What comes next

1. **Send the corrected work note and reply to Prof. Taniike.** Both are ready.
 2. **Run the prospective validation.** He offered to synthesise our candidates. This is the strongest path to publication.
 3. **Ask for the missing condition data.** Each catalyst is run at 5 temperatures under about 27 reaction-condition settings each. That is 135 runs per catalyst, and it matches the number Prof. Taniike quotes. We can see the structure in the row counts: 15 catalysts hold exactly 135 rows, and every one of them splits as exactly 27 rows at each of the 5 temperatures. But the file does not record what those 27 settings are. Because they are missing, 19.9% of the yield variance cannot be explained by composition and temperature alone. This is not noise. It is real chemistry we simply cannot see. Getting those columns would help more than any modelling change.
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11. What we can claim honestly

We can claim three things.

1. **A validation finding.** We documented a case where a literature-transfer method looked strong under standard cross-validation and failed under catalyst-grouped validation. We identified the mechanism. Random splits are common in this field, so this is worth reporting.
2. **A working screening tool.** The model ranks unseen catalysts well enough to guide synthesis.
3. **A scoping result.** Literature data does not help where our own coverage is good. It does help where our coverage is absent.
4. **A robustness result.** Our recommendation does not depend on an open question about how the data was collected. We tested it both ways and the shortlist barely moves.

We cannot claim that PFT improves prediction. We tested it properly, and it does not.